ARTIC-6 — Bill of Materials v0.3

> Last updated: March 2026
> **Prioritizing AliExpress for cost efficiency.** Shipping takes 2-4 weeks.
> Amazon alternatives listed where faster shipping is critical.

Changelog — v0.3

- Switched to AliExpress links where possible (30-50% cheaper)
- Added direct AliExpress search links
- Reduced parts cost from $985 → $622 (core parts only)
- Added realistic all-in cost section ($988-$1,424 including tools, shipping, tax, spares)

Changelog — v0.2 (corrections from v0.1)

Critical Fixes

- **NEMA 23 shaft diameter confirmed: 8mm** — specified STEPPERONLINE 23HS30-2804S with explicit 8mm shaft listing. Generic "NEMA 23 2.8A" listings often ship 6.35mm — do NOT order without checking.
- **GT2 80T pulley is aluminum, NOT printed** — found real aluminum 80T 8mm bore on Amazon (ASIN B08NXF2LWZ). Removed "(3D print)" note entirely. Printing the 80T driven pulley under 7Nm would destroy it.
- **Belt strategy changed: open-loop PU+steel-core, not closed-loop** — true closed-loop steel-core GT2 belts do not exist on Amazon in useful lengths. Closed-loop Amazon belts are rubber+fiberglass (more stretch). Correct approach: buy open-loop PU+steel-core white belt by the meter, cut to length, use GT2 belt clamps. Belt lengths still TBD after Fusion 360 center-distance is locked — **do NOT order belts until CAD is done.**
- **Standoff quantity corrected: 4→ 20** — Link 1 (4 standoffs), Link 2 (3 standoffs), base-to-turntable (4 standoffs), joint brackets (4 standoffs), spare (5). Previous BOM qty of 4 would result in a floppy sandwich.
- **51100 thrust bearing OD confirmed: 24mm** — CAD spec said 26mm seat. Correct seat recess: **24.5mm OD, 0.5mm deep**. Fix this in Fusion before DXF export.
- **TMC2209 sourcing: BIGTREETECH V1.3 only** — generic clone TMC2209s have UART pinout differences that will destroy them when wired to Arduino. BTT V1.3 is the verified safe choice.
- **NEMA 17 upsized: 1.5A → 2A, 59Ncm** — the wrist (Axes 4+5) is not as light as assumed when carrying a loaded gripper. Use the 2A 59Ncm STEPPERONLINE version for all 3 NEMA 17 slots.

MOTORS & DRIVERS

Qty	Part	Where to Buy	Specs	Each	Total

Qty	Part	Where to Buy	Specs	Each	Total
3	**NEMA 23 Stepper — 1.9Nm, 2.8A, 8mm shaft**	AliExpress	57×57×76mm, 8mm D-shaft, 4-wire bipolar. **VERIFY 8mm shaft before buying**	$12	$36
3	**NEMA 17 Stepper — 2A, 59Ncm, 5mm shaft**	AliExpress	42×42×48mm, 1.8°, 4-wire bipolar	$8	$24
8	**TMC2209 V1.3 Driver**	AliExpress	BTT or Fysetc brand. 2A cont, StallGuard4, UART. **Avoid generic clones**	$3	$24

> ⚠ **CRITICAL:** Verify NEMA 23 has 8mm shaft (not 6.35mm). Check listing photos for shaft diameter. BTT/Fysetc TMC2209 only — generic clones have wrong UART pinout.

Motors + Drivers Subtotal: $84 *(was $166 on Amazon)*

GEAR REDUCTION — ALL ALUMINUM — NO PRINTED PULLEYS

> 10:1 minimum on Axis 2 (shoulder) and Axis 3 (elbow).
> 20T (motor) → 80T (joint) = **4:1 per stage**. Two stages = 16:1 theoretical, ~10-12:1 effective.

Qty	Part	Where to Buy	Notes	Each	Total
4	**GT2 20T Pulley — 8mm bore, aluminum**	AliExpress	Fits NEMA 23 8mm shaft	$1.50	$6
2	**GT2 60T Pulley — 8mm bore, aluminum**	AliExpress	6mm belt width	$3	$6
2	**GT2 80T Pulley — 8mm bore, aluminum**	AliExpress	**Must be aluminum, NOT 3D printed**	$4	$8
5m	**GT2 Open Belt — PU + steel core, 6mm**	AliExpress	White PU, cut to length. **Order after CAD done**	$6	$6
1	**GT2 Belt Clamp Kit**	AliExpress	Copper buckles for open-loop ends	$3	$3

> ⚠ **Belt length:** Order after CAD is complete. Formula: Belt = 2×center_distance + π×(R_large + R_small)

Gear Reduction Subtotal: $29 *(was $56 on Amazon)*

COMPUTE & ELECTRONICS

Qty	Part	Where to Buy	Notes	Each	Total
1	**Raspberry Pi 5 — 4GB**	PiShop.us	Stock varies. Check PiShop first	$60	$60
1	**Arduino Mega 2560 Clone**	AliExpress	Get ATmega16U2 USB chip (not CH340)	$8	$8
1	**24V 10A DC PSU**	AliExpress	240W minimum	$12	$12
1	**5V 3A USB-C PSU (RPi)**	AliExpress	Dedicated for RPi5 only	$6	$6
1	**22mm E-Stop Button — NC**	AliExpress	Red mushroom, twist-release, **must be NC**	$3	$3
10	**KW11-3Z Limit Switch — SPDT**	AliExpress	Wire to NC terminal only	$0.30	$3
1	**8-Ch Opto-Isolated Relay**	AliExpress	Must be opto-isolated	$4	$4
1	**Logic Level Converter 4-ch**	AliExpress	3.3V ↔ 5V bidirectional	$1	$1
5	**1000µF 35V Capacitor**	AliExpress	One per NEMA 23 driver + spares	$0.20	$1
1	**Ferrite Core Kit**	AliExpress	Snap-on type	$2	$2

> ⚠ **Limit switch wiring:** KW11-3Z has COM, NO, NC pins. Wire COM→+5V, NC→Arduino pin with 10kΩ pulldown. Wire break = pin goes LOW = arm stops (fail-safe).

Compute + Electronics Subtotal: $100 *(was $165 on Amazon)*

STRUCTURE & HARDWARE

Qty	Part	Where to Buy	Notes	Each	Total
1	**SendCutSend — 3mm 6061-T6 Al**	sendcutsend.com	Base plate, 2× Link1, 2× Link2, brackets, mounts	—	$185
2	**51100 Thrust Bearing (10×24×9mm)**	AliExpress	10mm bore, 24mm OD. 1 spare	$2	$4
20	**M5 × 40mm Aluminum Standoffs**	AliExpress	Hex, male-female	$0.80	$16
1	**M3/M4/M5 Bolt & Nut Kit (500pc)**	AliExpress	Socket head cap screws	$8	$8
1	**Brass Heat-Set Insert Kit**	AliExpress	M3/M4 for PETG parts	$5	$5
20	**608ZZ Bearings**	AliExpress	8×22×7mm. Buy 20, need 12	$0.25	$5
6	**8mm Rigid Shaft Coupler**	AliExpress	Aluminum	$1.50	$9

Qty	Part	Where to Buy	Notes	Each	Total
1	**Fender Washer Kit — M3/M4/M5**	AliExpress	Under bolts touching PETG	$4	$4

> ⚠ **Standoff placement:** Every 100mm minimum. A 350mm link with only end standoffs will bow.

Structure + Hardware Subtotal: $236 *(was $301 on Amazon — SendCutSend same price)*

VISION & SENSING

Qty	Part	Where to Buy	Notes	Each	Total
1	**Intel RealSense D435 — used**	eBay	Target $80-100 used. Check USB-C port	$90	$90
1	**1kg Load Cell + HX711**	AliExpress	Tare every grip cycle	$3	$3
1	**USB Microphone**	AliExpress	Must be USB (RPi5 has no analog audio)	$8	$8

Vision + Sensing Subtotal: $101 *(was $130)*

FILAMENT

> PETG: covers, housings, wrist assembly, gripper body only. No structural PETG near motors.

Qty	Part	Where to Buy	Notes	Each	Total
3	**PETG Filament — 1kg**	AliExpress	230-240°C print temp	$12	$36
1	**TPU 95A Filament — 500g**	AliExpress	Gripper fingertips only	$8	$8

Filament Subtotal: $44 *(was $82)*

WIRING & CABLE MANAGEMENT

Qty	Part	Where to Buy	Notes	Each	Total
1	**Silicone Wire Kit — 22/26AWG**	AliExpress	High-flex, ≥200°C rated	$6	$6
5m	**Shielded Twisted Pair — 24AWG**	AliExpress	For limit switches	$4	$4
1	**JST-XH Connector Kit**	AliExpress	2.54mm pitch	$4	$4
1	**Cable Drag Chain — 10×15mm, 1m**	AliExpress	Check bend radius	$4	$4
1	**Braided Cable Sleeve Kit**	AliExpress		$3	$3
1	**Heat Shrink Tubing Kit**	AliExpress		$2	$2

Qty	Part	Where to Buy	Notes	Each	Total
1	**Copper Ground Bus Bar**	AliExpress	Star ground point	$3	$3
1	**12-Way Terminal Block**	AliExpress	Power distribution	$2	$2

Wiring Subtotal: $28 *(was $85)*

REALISTIC ALL-IN COST

> **This is the honest total** — including tools, shipping, tax, consumables, and contingency. Most BOMs hide these. I'm not hiding anything.

Parts List (Core Components)

Category	Low Est.	High Est.
Core Parts (Motors, Electronics, Structure)	$580	$640
3D Printing Filament (PETG + TPU)	$60	$82
SendCutSend Aluminum Cutting	$110	$150
Parts Subtotal	**$750**	**$872**

Hidden & Missed Costs (What Most BOMs Don't Show)

Category	Low Est.	High Est.
Shipping — AliExpress	$18	$35
Shipping — Amazon	$0	$20
Sales Tax (~8%)	$45	$70
Tools (soldering iron, multimeter, crimpers)	$0	$132
Consumables (solder, flux, threadlocker, grease)	$33	$58
RPi Essentials (SD card, cooler, USB hub)	$30	$48
Prototyping (breadboard, jumpers, caps)	$17	$28
Installation (desk mount, velcro)	$14	$25
Hidden Costs Subtotal	**$157**	**$416**

Spares & Contingency

Item	Low Est.	High Est.
Spare NEMA 17 Motor (1x)	$12	$12
Spare TMC2209 Driver (2x)	$14	$14

Item	Low Est.	High Est.
DOA / Wrong Parts	$20	$40
Failed Prints / Wasted Filament	$15	$30
Oops Fund	$20	$40
Spares & Contingency Subtotal	**$81**	**$136**

💰 GRAND TOTAL — REALISTIC

	Low	High
Parts List (accounted)	$750	$872
Hidden / Missed Costs	$157	$416
Spares & Contingency	$81	$136
GRAND TOTAL	**$988**	**$1,424**

> If you own tools already (soldering iron, multimeter, wire strippers): subtract ~$100. Realistic midpoint estimate: **~$1,100-$1,200**.

Phased Buying Plan

Phase	What	Cost	When
1	Fusion 360 Design	FREE	Month 1
2	Structure & Motors	$434	Month 1-2
3	Electronics & Power	$147	Month 2
4	Brain & Senses	$268	Month 2-3
5	Tools, Consumables, Spares	$223	Ongoing
	TOTAL	**$1,072**	

Blueprint Grant Request: $350

I'm not asking for the full $1,100. I'm asking for $350 to cover Phase 2 motion components.

What Grant Covers	Cost
3× NEMA 23 Steppers	$36
3× NEMA 17 Steppers	$24
6× TMC2209 Drivers	$24
GT2 Pulleys + Belt	$42

What Grant Covers	Cost
Bolts, bearings, standoffs	$42
Wiring supplies	$28
Shipping (AliExpress)	$35
Buffer for price changes	$119
Grant Request	**$350**

What I'm Self-Funding

What I Pay For	Cost
SendCutSend aluminum	$130
Raspberry Pi 5 (4GB)	$60
Intel RealSense D435 (used)	$95
RPi essentials (SD, cooler, hub)	$46
24V PSU + Arduino + E-stop	$57
Filament (PETG + TPU)	$82
All tools (soldering, etc.)	$55
All consumables	$58
Spares & contingency	$136
My Contribution	**~$719**

> **Total project: ~$1,100 | Grant: $350 | Self-funded: ~$750** I'm covering 68% of the project myself.